Supplementary Material for "System Relaxation for Resilient Network Governance: A Model-Driven Control Mechanism Against Structural Stress Concentration" (Manuscript ID: CNCOMM-2026-0222)

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This supplement gives the complete formal proofs. We use $X = [0,1]^N$, $F(S) = (1-\alpha)S + \alpha\mathcal{N}(S)$, $0 < \alpha < 1$, where $\mathcal{N}$ is the Krasovskii regularization of the normalized routing stress. Equation numbers cited in prose refer to the main manuscript.

S-A Proof of Lemma 1

Proof. (a) Positive Invariance: Take any $S^{(k)} \in X$. By max-value normalization, $\mathcal{N}(S^{(k)}) \subset [0,1]^N = X$. For any choice of $c^{(k)} \in \mathcal{N}(S^{(k)})$ and any coordinate i, we have

$$S_i^{(k+1)} = (1-\alpha)S_i^{(k)} + \alpha\, c_i^{(k)}, \quad \text{where} \quad S_i^{(k)}, c_i^{(k)} \in [0,1]. \tag{1}$$

Thus,

$$0 \cdot (1-\alpha) + 0 \cdot \alpha \;\le\; S_i^{(k+1)} \;\le\; 1 \cdot (1-\alpha) + 1 \cdot \alpha, \tag{2}$$

which implies $S_i^{(k+1)} \in [0,1]$. As this holds for each coordinate, $S^{(k+1)} \in X$. This shows that $F(X) \subset X$: **if the state is in X at the current step, it remains in X at the next step**. Therefore, X is a positively invariant set.

(b) Compactness: The set $X = [0,1]^N$ is a closed and bounded hypercube in $\mathbb{R}^N$, and is therefore compact.

(c) Forward Invariance on the Effective State Space: The algorithm and analysis in this paper are initialized within X ($S^{(0)} \in X$), and from (a), we know $F(X) \subset X$. Therefore, for the discrete-time semi-flow on the state space X,

$$\Phi(k,\cdot) : X \to 2^X, \quad \Phi(1,\cdot) = F, \;\; \Phi(k+1,\cdot) = F \circ \Phi(k,\cdot), \tag{3}$$

any bounded subset $B \subset X$ remains in X for all $n \ge 0$ ($\Phi(n,B) \subset X$).

(d) Exponential Attraction from Perturbed Initial States $S^{(0)} \in [-1/\beta_I + \epsilon, \infty)^N$: Suppose a perturbed initial state $S^{(0)}$ lies outside $X = [0,1]^N$ while preserving positive node costs $C_i^{(0)} = 1 + \beta_I S_i^{(0)} > 0$. For each step $k \ge 0$ and coordinate $i \in \{1,\dots,N\}$, define the coordinate-wise upper and lower box-violation excesses with respect to $[0,1]$:

$$\delta_{k,i}^{+} := \max\{S_i^{(k)} - 1,\, 0\}, \qquad \delta_{k,i}^{-} := \max\{-S_i^{(k)},\, 0\}. \tag{4}$$

Since $c_i^{(k)} \in [0,1]$, we have $S_i^{(k+1)} - 1 = (1-\alpha)(S_i^{(k)} - 1) + \alpha(c_i^{(k)} - 1) \le (1-\alpha)(S_i^{(k)} - 1)$ and $-S_i^{(k+1)} = (1-\alpha)(-S_i^{(k)}) - \alpha c_i^{(k)} \le (1-\alpha)(-S_i^{(k)})$. Taking the positive part yields

$$\delta_{k+1,i}^{+} \;\le\; (1-\alpha)\,\delta_{k,i}^{+}, \qquad \delta_{k+1,i}^{-} \;\le\; (1-\alpha)\,\delta_{k,i}^{-}. \tag{5}$$

This means that $\max\{\delta_{k,i}^+, \delta_{k,i}^-\}$ converges geometrically to 0 at a rate of $1-\alpha$, and thus

$$\operatorname{dist}_\infty(S^{(k)}, X) \le (1-\alpha)^k \operatorname{dist}_\infty(S^{(0)}, X) \to 0. \tag{6}$$

This shows **exponential attraction** to X. Since Algorithm 1 initializes $S^{(0)} = \mathbf{0} \in X$, all trajectories remain strictly inside X for all $k \ge 0$.

In conclusion, $X = [0,1]^N$ is positively invariant, compact, and exponentially attracting for the system. □

S-B Proof of Proposition 1

Proof. (a) Non-empty, Compact, and Convex Values: By definition of the Krasovskii regularization in Eq. (7), for any $S \in X$,

$$\mathcal{N}(S) := \bigcap_{\delta>0} \overline{\operatorname{co}}\big(\operatorname{norm}(\kappa(B_\delta(S)))\big) \subset [0,1]^N. \tag{7}$$

Because $\operatorname{norm}(\kappa(S')) \in [0,1]^N$ for every $S' \in B_\delta(S)$, each set $C_\delta(S) := \overline{\operatorname{co}}\big(\operatorname{norm}(\kappa(B_\delta(S)))\big)$ is a non-empty, closed, bounded, and convex subset of the compact hypercube $[0,1]^N$. As the nested intersection of non-empty compact convex sets as $\delta \downarrow 0$, $\mathcal{N}(S)$ is **non-empty, compact, and convex**.

(b) Piecewise Constant Structure and Finiteness: The graph G and the set of source-destination (OD) pairs are finite. The node cost $C(S)$ is an affine function of S. For each OD pair (s,t), the set of candidate simple paths is **finite**. The cost of each path is an affine function of S. Thus, there exists a **finite polyhedral decomposition** of the state space X (comprising full-dimensional open polyhedral cells and lower-dimensional tie-breaking faces) such that within the relative interior of each stratum, the set of active shortest paths for every OD pair remains **constant**. Therefore, within each stratum $\ell \in \{1,\dots,M\}$, $\kappa(S)$ (as well as $\widetilde{\kappa}(S;W,\mu)$) and $\operatorname{norm}(\kappa(S)) \equiv v_\ell$ are **constant vectors**. For any point $S \in X$, let $\mathcal{I}(S) := \{\ell \in \{1,\dots,M\} : S \in \overline{\mathcal{R}_\ell}\}$ denote the finite index set of strata whose closures contain S. Then for all sufficiently small $\delta > 0$, $B_\delta(S)$ intersects exactly the strata in $\mathcal{I}(S)$, so

$$\mathcal{N}(S) = \operatorname{co}\big(\{v_\ell :\ \ell \in \mathcal{I}(S)\}\big). \tag{8}$$

This establishes the structure of $\mathcal{N}(\cdot)$ as being **piecewise constant on open cells with the closed convex hull of incident strata taken on boundary faces**.

(c) Upper Semi-Continuity (USC) via Sequential Criterion / Closed Graph: Take an arbitrary sequence $S_n \to S$, and let $c_n \in \mathcal{N}(S_n)$ such that $c_n \to c$. We must show that $c \in \mathcal{N}(S)$. From (b), for each n, there exists a finite set $\{v_\ell : \ell \in \mathcal{I}(S_n)\}$ such that $c_n \in \operatorname{co}\big(\{v_\ell : \ell \in \mathcal{I}(S_n)\}\big)$. By Carathéodory's theorem, we can write

$$c_n = \sum_{j=0}^{N} \lambda_{n,j}\, v_{\ell_{n,j}}, \quad \lambda_{n,j} \ge 0,\ \sum_{j=0}^{N} \lambda_{n,j} = 1,\ \ell_{n,j} \in \mathcal{I}(S_n). \tag{9}$$

Since the vectors v_ℓ come from a **globally finite** set $\{v_1,\dots,v_M\}$, we can pass to a subsequence to ensure $\ell_{n,j} \equiv \bar{\ell}_j$ is **fixed**, and (by compactness of $[0,1]$) $\lambda_{n,j} \to \bar{\lambda}_j$. Then,

$$c = \lim_{n\to\infty} c_n = \sum_{j=0}^{N} \bar{\lambda}_j\, v_{\bar{\ell}_j}, \quad \sum_{j=0}^{N} \bar{\lambda}_j = 1,\ \bar{\lambda}_j \ge 0. \tag{10}$$

Since $S_n \to S$, for n large enough, any limit index $\bar{\ell}_j$ must belong to $\mathcal{I}(S)$. Thus, each limit vector $v_{\bar{\ell}_j}$ is one of the constant vectors from a stratum incident to S. Therefore,

$$c = \sum_{j=0}^{N} \bar{\lambda}_j \, v_{\bar{\ell}_j} \;\in\; \mathrm{co}\big(\{v_\ell : \ell \in \mathcal{I}(S)\}\big) = \mathcal{N}(S). \tag{11}$$

This means the graph of $\mathcal{N}$, $\mathrm{gph}\mathcal{N} := \{(S, c) : c \in \mathcal{N}(S)\}$, is closed. Since $\mathcal{N}$ maps into the compact set X, a closed-graph map is **USC** [1].

(d) Properties of $F(\cdot)$: Scaling and translation preserve the USC property and non-empty, compact, convex values. The map $S \mapsto (1-\alpha)S$ is continuous (single-valued). The Minkowski sum $F(S) = (1-\alpha)S + \alpha\mathcal{N}(S)$ of a continuous map and a USC map is also USC with non-empty, compact, and convex values. □

S-C Proof of Theorem 1

Proof. (a) Nested Sequence of Compact Forward Images: Define the forward image sequence $K_0 := X = [0,1]^N$ and $K_{n+1} := F(K_n) = \bigcup_{S \in K_n} F(S)$ for $n \geq 0$. By Lemma 1, $K_1 = F(X) \subset X = K_0$. By induction, $K_{n+1} \subset K_n$ for all $n \geq 0$. Because F is upper semi-continuous with non-empty compact values (Proposition 1) and the image of a compact set under a compact-valued USC map is compact [1], each K_n is a non-empty compact subset of X.

(b) Non-Emptiness and Compactness of Ω: By Cantor's intersection theorem, the nested intersection of non-empty compact sets

$$\Omega := \bigcap_{n=0}^{\infty} K_n = \bigcap_{n=0}^{\infty} F^n(X) \tag{12}$$

is **non-empty** and **compact**.

(c) Invariance ($F(\Omega) = \Omega$): Since $\Omega \subset K_n$ for every n, we have $F(\Omega) \subset F(K_n) = K_{n+1}$ for all $n \geq 0$, which implies $F(\Omega) \subset \bigcap_{n\geq 0} K_{n+1} = \Omega$ (positive invariance). Conversely, take any $y \in \Omega$. Then for each $n \geq 1$, $y \in K_n = F(K_{n-1})$, so there exists $x_n \in K_{n-1}$ such that $y \in F(x_n)$. Since X is compact, $\{x_n\}$ has a convergent subsequence $x_{n_j} \to x^\star$. Because K_m is closed and $x_{n_j} \in K_m$ for all $n_j > m$, the limit satisfies $x^\star \in \bigcap_{m\geq 0} K_m = \Omega$. By the closed-graph property of F (Proposition 1), $x_{n_j} \to x^\star$ and $y \in F(x_{n_j})$ imply $y \in F(x^\star) \subset F(\Omega)$. Thus $\Omega \subset F(\Omega)$, establishing exact invariance $F(\Omega) = \Omega$.

(d) Global Uniform Attraction on X: For any initial state $S^{(0)} \in X$, the k-th iterate satisfies $S^{(k)} \in F^k(S^{(0)}) \subset F^k(X) = K_k$. Suppose for contradiction that $\sup_{S^{(0)}\in X} \mathrm{dist}(S^{(k)}, \Omega)$ does not converge to 0. Then there exist $\epsilon > 0$ and a sequence $z_{k_j} \in K_{k_j}$ ($k_j \uparrow \infty$) such that $\mathrm{dist}(z_{k_j}, \Omega) \geq \epsilon$. By compactness of X, a subsequence converges to some $z^\star \in X$. Since K_m is closed and $z_{k_j} \in K_{k_j} \subset K_m$ for all $k_j \geq m$, we have $z^\star \in \bigcap_{m\geq 0} K_m = \Omega$, contradicting $\mathrm{dist}(z^\star, \Omega) \geq \epsilon$. Hence, for every trajectory starting in X,

$$\lim_{k\to\infty} \mathrm{dist}\big(S^{(k)}, \Omega\big) = 0, \tag{13}$$

and the ω-limit set of every trajectory initialized in X satisfies $\emptyset \neq \omega(S^{(0)}) \subset \Omega$. □

S-D Proof of Theorem 2

Proof. At step k, the SRA controller updates the pressure state via $S^{(k+1)} = (1-\alpha)S^{(k)} + \alpha\, q^{(k)}$, where $q^{(k)} = \mathrm{norm}(\kappa^{(k)}) \in \{v_1, \ldots, v_M\}$ is the active normalized stress vector, and $E_k = \|S^{(k)} -$

$q^{(k)}\|_2^2$ is the tracking energy. Subtracting the next-step active stress vector $q^{(k+1)}$ from both sides yields the exact algebraic identity:

$$S^{(k+1)} - q^{(k+1)} = (1-\alpha)\big(S^{(k)} - q^{(k)}\big) + \big(q^{(k)} - q^{(k+1)}\big). \tag{14}$$

Notice first that if no path switch occurs between steps k and $k+1$, then $q^{(k+1)} = q^{(k)}$, so $E_{k+1} = \|S^{(k+1)} - q^{(k+1)}\|_2^2 = (1-\alpha)^2\|S^{(k)} - q^{(k)}\|_2^2 = (1-\alpha)^2 E_k \le (1-\alpha)E_k$.

In the general case where a path switch may occur ($q^{(k+1)} \neq q^{(k)}$), we apply the Weighted Young's Inequality $\|x+y\|_2^2 \le (1+\lambda)\|x\|_2^2 + (1+1/\lambda)\|y\|_2^2$ with $x = (1-\alpha)(S^{(k)} - q^{(k)})$, $y = q^{(k)} - q^{(k+1)}$, and weight $\lambda = \frac{\alpha}{1-\alpha} > 0$. Since $(1+\lambda)(1-\alpha)^2 = 1-\alpha$ and $1 + \frac{1}{\lambda} = \frac{1}{\alpha}$, we obtain

$$E_{k+1} = \|S^{(k+1)} - q^{(k+1)}\|_2^2 \le (1-\alpha)E_k + \frac{1}{\alpha}\|q^{(k)} - q^{(k+1)}\|_2^2. \tag{15}$$

Since $q^{(k)}, q^{(k+1)} \in \mathrm{co}(\{v_1, \dots, v_M\})$ whose diameter is $\bar{\Delta} = \max_{\ell,m} \|v_\ell - v_m\|_2$, we have $\|q^{(k)} - q^{(k+1)}\|_2 \le \bar{\Delta}$, yielding Eq. (11):

$$E_{k+1} \le (1-\alpha)E_k + \frac{\bar{\Delta}^2}{\alpha}. \tag{16}$$

Unrolling this linear recurrence gives $E_k \le (1-\alpha)^k E_0 + \frac{\bar{\Delta}^2}{\alpha^2}$. Taking $\limsup_{k\to\infty}$ yields Eq. (12). Moreover, at every recomputation epoch where $q^{(k)} \in \mathcal{N}(S^{(k)})$, $V(S^{(k)}) = \min_{c\in\mathcal{N}(S^{(k)})} \|S^{(k)} - c\|_2^2 \le E_k$ satisfies the same asymptotic bound. □

S-E Proof of Corollary 1

Proof. (1) **Uniqueness of Fixed Point:** Solving $S = (1-\alpha)S + \alpha c$ gives $\alpha S = \alpha c$, so $S = c$. The unique fixed point is $S^\star = c$. (2) **Contraction and Convergence Rate:**

$$S^{(k+1)} - c = (1-\alpha)S^{(k)} + \alpha c - c = (1-\alpha)\big(S^{(k)} - c\big). \tag{17}$$

Taking the 2-norm and iterating gives the result:

$$\|S^{(k)} - S^\star\|_2 = (1-\alpha)^k \|S^{(0)} - S^\star\|_2. \tag{18}$$

□

S-F Proof of Proposition 2

Proof. (i) Within dwell block m, the active path set is held fixed for $t \in \{0, \dots, T_d - 1\}$, so $q^{(\tau_m + t)} \equiv q^{(\tau_m)}$ is constant. Consequently, for $t \in \{0, \dots, T_d - 1\}$, $S^{(\tau_m+t)} - q^{(\tau_m)} = (1-\alpha)^t (S^{(\tau_m)} - q^{(\tau_m)})$, and taking the squared Euclidean norm yields

$$E_{\tau_m + t} = (1-\alpha)^{2t} E_{\tau_m} = \gamma^{2t} V_m \le \gamma^t V_m. \tag{19}$$

(ii) At the last step of block m (transitioning from $\tau_m + T_d - 1$ to $\tau_{m+1} = \tau_m + T_d$), a path switch is permitted ($q^{(\tau_{m+1})}$ may differ from $q^{(\tau_m + T_d - 1)} = q^{(\tau_m)}$). Applying the one-step Drop-Jump inequality from Theorem 2 at $k = \tau_{m+1} - 1$ gives:

$$\begin{aligned} V_{m+1} = E_{\tau_{m+1}} &\le \gamma\, E_{\tau_{m+1}-1} + \tilde{C} \\ &= \gamma\, E_{\tau_m + T_d - 1} + \tilde{C} \\ &\le \gamma \cdot (\gamma^{T_d - 1} V_m) + \tilde{C} = \gamma^{T_d} V_m + \tilde{C}, \end{aligned} \tag{20}$$

where the last inequality uses the intra-block cooling bound from (i). □

S-G Proof of Theorem 3

Proof. (a) Iteratively expanding the linear recurrence from Proposition 2(ii), $V_{m+1} \le \gamma^{T_d} V_m + \tilde{C}$, gives:

$$\begin{aligned} V_m &\le (\gamma^{T_d})^m V_0 + \tilde{C} \sum_{j=0}^{m-1} (\gamma^{T_d})^j \\ &= \gamma^{mT_d} V_0 + \tilde{C}\, \frac{1-\gamma^{mT_d}}{1-\gamma^{T_d}} \\ &\le \gamma^{mT_d} V_0 + \frac{\tilde{C}}{1-\gamma^{T_d}}. \end{aligned} \tag{21}$$

Taking the limit superior as $m \to \infty$ gives the steady-state radius. (b) If $V_0 \le \varepsilon^2 - \frac{\tilde{C}}{1-\gamma^{T_d}}$ (including $V_0 = 0$), then $V_m \le \varepsilon^2$ holds from $m = 0$. For $V_0 > \varepsilon^2 - \frac{\tilde{C}}{1-\gamma^{T_d}} > 0$, setting $\gamma^{mT_d} V_0 + \frac{\tilde{C}}{1-\gamma^{T_d}} \le \varepsilon^2$ and solving for m yields Eq. (17). □

S-H Proof of Theorem 4

Proof. (a) By Algorithm 1, shortest-path recomputations and route switches occur only at block boundaries $\tau_m = mT_d$, separated by $T_d - 1$ frozen steps. Thus, the number of switches in any window of length $T \ge 1$ satisfies $N_{\mathrm{sw}}(T) \le \lceil T/T_d \rceil$, which implies $\bar{f}_{\mathrm{sw}} = \limsup_{T\to\infty} N_{\mathrm{sw}}(T)/T \le 1/T_d$. (b) This result is a direct consequence of solving the inter-block recurrence, as shown in the proof of Theorem 3(a), where $\tilde{C}/(1-\gamma^{T_d})$ decreases monotonically toward $\tilde{C}$ with T_d. □

S-I Proof of Lemma 2

Proof. The proof proceeds in two direct steps: (1) bounding the post-switch stress of the relieved peak node u and all non-peak nodes $j \ne u$, and (2) taking the maximum over V.
1) Route-Redistribution Budget and Node-wise Bounds: Let $\kappa^{(m)}$ and $\kappa^{(m+1)}$ denote the stress profiles before and after the switching event between blocks m and $m+1$, and let u be the unique pre-switch peak node with $\kappa_u^{(m)} = \kappa_{\max}^{(m)} > \kappa_j^{(m)}$ $(\forall j \ne u)$. Under the positive load-shedding hypothesis $\Delta_u^- := \kappa_u^{(m)} - \kappa_u^{(m+1)} \ge \delta > 0$, the post-switch load at node u satisfies

$$\kappa_u^{(m+1)} = \kappa_{\max}^{(m)} - \Delta_u^- \le \kappa_{\max}^{(m)} - \delta. \tag{22}$$

Meanwhile, for any other node $j \in V \setminus \{u\}$, let $\mathcal{D}_{\mathrm{in}}^{(m)}(j)$ denote the set of OD pairs whose shortest-path transit fraction $\sigma_{st}(j)/\sigma_{st}$ through node j increases at step $m+1$. The net stress increment on node j is bounded by its route-redistribution budget $B_j^{(m)}$:

$$\kappa_j^{(m+1)} - \kappa_j^{(m)} \le (\kappa_j^{(m+1)} - \kappa_j^{(m)})_+ =: B_j^{(m)} \le \sum_{(s,t)\in\mathcal{D}_{\mathrm{in}}^{(m)}(j)} W_{st} \tag{23}$$

(where $W_{st} \equiv 1$ for unweighted BC, so $B_j^{(m)} \le |\mathcal{D}_{\mathrm{in}}^{(m)}(j)|$).
2) Strict Peak Reduction under Non-Overloading Diversion: By the Non-Overloading Diversion Condition in Eq. (19), every node $j \in V \setminus \{u\}$ has sufficient slack to absorb $B_j^{(m)}$ without exceeding $\kappa_{\max}^{(m)} - \delta$:

$$\kappa_j^{(m+1)} \le \kappa_j^{(m)} + B_j^{(m)} \le \kappa_{\max}^{(m)} - \delta, \qquad \forall j \in V \setminus \{u\}. \tag{24}$$

Combining the bound for $i = u$ and the bound for all $j \in V \setminus \{u\}$ yields

$$\kappa_{\max}^{(m+1)} = \max\left\{\kappa_u^{(m+1)}, \max_{j \in V\setminus\{u\}} \kappa_j^{(m+1)}\right\} \le \kappa_{\max}^{(m)} - \delta, \tag{25}$$

completing the proof. □

S-J Proof of Proposition 3

Proof. The proof establishes each of the three claims using the peak reduction $\kappa'_{\max} \le \kappa_{\max} - \delta < \kappa_{\max}$ from Lemma 2 (with $d := \Delta_u^- \ge \delta > 0$).
(i) Unconditional Expansion of Robust Capacity Lower Bound: For any node service rate vector $\mu = (\mu_i)_{i\in V}$ with $\mu_{\min} = \min_{i\in V} \mu_i > 0$, the actual maximum stable traffic multiplier satisfies

$$\rho_{\max}(\kappa) = \min_{i\in V} \frac{\mu_i}{\kappa_i} \ge \frac{\min_{i\in V} \mu_i}{\max_{i\in V} \kappa_i} = \frac{\mu_{\min}}{\kappa_{\max}} = L(\kappa). \tag{26}$$

Since $\mu_{\min} > 0$ is invariant under routing updates and $\kappa'_{\max} < \kappa_{\max}$, we immediately have

$$L(\kappa') = \frac{\mu_{\min}}{\kappa'_{\max}} > \frac{\mu_{\min}}{\kappa_{\max}} = L(\kappa). \tag{27}$$

(ii) Exact Capacity Proportionality under Homogeneous Service Rates or Generalized Stress: When node service rates are homogeneous ($\mu_i \equiv \mu > 0$ for all $i \in V$), factoring out the constant μ yields the exact identity:

$$\rho_{\max}(\kappa) = \min_{i\in V} \frac{\mu}{\kappa_i} = \frac{\mu}{\max_{i\in V} \kappa_i} = \frac{\mu}{\kappa_{\max}}, \qquad \rho_{\max}(\kappa') = \min_{i\in V} \frac{\mu}{\kappa'_i} = \frac{\mu}{\kappa'_{\max}}. \tag{28}$$

Dividing the two expressions gives the exact proportionality in Eq. (22):

$$\frac{\rho_{\max}(\kappa')}{\rho_{\max}(\kappa)} = \frac{\mu/\kappa'_{\max}}{\mu/\kappa_{\max}} = \frac{\kappa_{\max}}{\kappa'_{\max}} > 1. \tag{29}$$

Similarly, under heterogeneous service rates $\mu_i > 0$ and demands $W_{st} \ge 0$, using the generalized stress $\widetilde{\kappa}_i(S; W, \mu) = \frac{1}{\mu_i} \sum_{s\ne i\ne t} W_{st} \frac{\sigma_{st}(i)}{\sigma_{st}}$ in Eq. (2) gives node i's utilization as $\rho\,\widetilde{\kappa}_i < 1$, so $\rho_{\max}(\widetilde{\kappa}) = \min_{i\in V}(1/\widetilde{\kappa}_i) = 1/\widetilde{\kappa}_{\max}$, and whenever $\widetilde{\kappa}'_{\max} < \widetilde{\kappa}_{\max}$ we obtain $\rho_{\max}(\widetilde{\kappa}')/\rho_{\max}(\widetilde{\kappa}) = \widetilde{\kappa}_{\max}/\widetilde{\kappa}'_{\max} > 1$.
(iii) Sufficient Condition under Heterogeneous Service Rates with Unweighted BC: Let $\rho_0 := \rho_{\max}(\kappa) = \min_{i\in V}(\mu_i/\kappa_i)$. Because $\rho_{\max}(\kappa') = \min_{i\in V}(\mu_i/\kappa'_i)$ is the minimum over all $i \in V$, the inequality $\rho_{\max}(\kappa') \ge \rho_0$ holds if and only if $\mu_i/\kappa'_i \ge \rho_0$ holds simultaneously for every $i \in V$. Since $\kappa'_u \le \kappa_{\max} - d$ for the relieved peak u and $\kappa'_j \le \kappa_j + B_j$ for all $j \in V\setminus\{u\}$, we have $\frac{\mu_u}{\kappa'_u} \ge \frac{\mu_u}{\kappa_{\max}-d}$ and $\frac{\mu_j}{\kappa'_j} \ge \frac{\mu_j}{\kappa_j+B_j}$. Consequently, the joint condition in Eq. (23):

$$\frac{\mu_u}{\kappa_{\max} - d} \ge \rho_0 \quad \text{and} \quad \frac{\mu_j}{\kappa_j + B_j} \ge \rho_0, \quad \forall j \in V \setminus \{u\}, \tag{30}$$

is sufficient to guarantee $\rho_{\max}(\kappa') = \min\left\{\frac{\mu_u}{\kappa'_u}, \min_{j\ne u} \frac{\mu_j}{\kappa'_j}\right\} \ge \rho_0$ (and strict inequality $\rho_{\max}(\kappa') > \rho_0$ whenever the inequalities in (23) are strict). □

References

[1] J.-P. Aubin and A. Cellina, *Differential Inclusions*. Springer, 1984.